

Stage 9 End of unit 9

The tests and mark schemes have been written by the authors. These may not fully reflect the approach of Cambridge Assessment International Education.

Answers

- 1** **a** 1st term: 1, 2nd term: 3, 3rd term: 11, 4th term: 123 [2]
b 1st term: 6, 2nd term: 9, 3rd term: 36, 4th term: 1089 [2]
c 1st term: 4, 2nd term: 5, 3rd term: 8, 4th term: 13 [2]
d 1st term: 22, 2nd term: 20, 3rd term: 16, 4th term: 10 [2]
- 2** **a** $2, 3\frac{3}{5}, 5\frac{1}{5}, 6\frac{4}{5}, 8\frac{2}{5}, 10, 11\frac{3}{5}$ [2]
b 25.6, 22.4, 19.2, 16, 12.8, 9.6, 6.4, 3.2 [2]
- 3** 4 [3]
- 4** **a** 1st term: -2, 2nd term: 1, 3rd term: 4, 10th term: 25 [2]
b 1st term: $2\frac{1}{4}$, 2nd term: $2\frac{1}{2}$, 3rd term: $2\frac{3}{4}$, 10th term: $4\frac{1}{2}$ [2]
c 1st term: 6, 2nd term: 13, 3rd term: 32, 10th term: 1005 [2]
- 5** **a** n th term is: $n^2 + 5$ [2]
b n th term is: $n^2 - 7$ [2]
- 6** **a** **A** term: $\frac{9}{12} = \frac{3}{4}$; **B** term: $\frac{6}{9} = \frac{2}{3}$; **C** term: $\frac{15}{18} = \frac{5}{6}$ [3]
b **B, A, C** or $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$ [1]
- 7** Yes, 15th term is $15^2 - 21 = 225 - 21 = 204$ [2]

8 a i

x	1	$1\frac{1}{2}$	± 3	$\pm 4\frac{1}{2}$
y	6	$7\frac{1}{4}$	14	$25\frac{1}{4}$

[2]

ii

x	-1	2	5	10
y	-9	0	117	992

[2]

b i $y = x^2 + 5$

ii $y = x^3 - 8$

[2]

9 a i $y = x^2 + 2$

ii $x = \pm\sqrt{y-2}$

[2]

b i $y = (x - 5)^2$

ii $x = \pm\sqrt{y} + 5$

[2]

c i $y = (3x)^3$

ii $x = \frac{\sqrt[3]{y}}{3}$

[2]

d i $y = \left(\frac{x}{4}\right)^2$

ii $x = \pm 4\sqrt{y}$

[2]

[Total: 45 marks]